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MS984: Data Analytics in Practice

IMPROVING EMERGENCY CARE ACCESS:
STRATEGIES FOR REDUCING A&E WAIT TIMES IN
SCOTLAND

Group 7



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1.0 Introduction

Recent reports (BMA Scotland, 2023) and surveys (UNISON Scotland, 2024) indicate that there are concerns about insufficient staffing levels and a shortage of doctors in the Scottish NHS. They have highlighted these issues, citing potential risks to patient care and increased stress for healthcare workers.

However, the Scottish government has implemented measures like the Health and Care (Staffing) (Scotland) Act 2019 to address these concerns and ensure safe staffing levels. It's important to note that while there are challenges, the Scottish government and healthcare organizations are working to address the doctor shortage and the consistency in quality of care.

In this section, it will dive deeper into two challenges that the NHS face, insufficient staffing level and the consistency in quality of care. By focusing on these two factors, it addresses both the short-term and long-term needs of healthcare systems. Ensuring the healthcare system could be effective and sustainable. Ultimately improving patient outcomes and overall health of Scotland.

2.0 Key Challenges

In Appendix 1 “Number of patients waiting for treatment” – the graph highlights a sharp increase in patients waiting for treatment post COVID-19, indicating the long-term stress the pandemic has placed on healthcare services. Addressing this backlog will most likely require additional resources and strategic planning.

In Appendix 2 “Waiting times in A&E” shows a similar trend to Appendix 1. The target of 95% was reached soon after the pandemic started, however, there is a dramatic decrease, with the number stabilising around 70% in 2024. Again, showcasing that the pandemic had an overall negative impact for the patients and the NHS.

2.1 Insufficient Staffing Level

The NHS in Scotland has been facing severe staffing crisis that threatens the status quo of providing timely and high-quality care. The issue has been exemplified by the COVID-19 pandemic, which created an unpredictable demand on the healthcare system. However, the root of this crisis goes beyond the pandemic, involving factors such as workforce planning problems, funding issues, recruitment and retention challenges.

One significant driver of staffing shortages is the increasing demand. The COVID-19 pandemic has caused an increase to delayed treatments and appointments, creating an ever-growing backlog that demands additional staffing to resolve.

In Appendix 3 “Growing Patient Load per GP” provides a visual of the ever-increasing burden on GPs. The graph shows the steady growth of number of patients per GP

since 2013, highlighting a notable 13% increase by 2023. The growing patient load can contribute to GP burnout. The constant pressure to meet demands leads to stress and a decline in job satisfaction. Addressing this issue needs efforts to increase the GP workforce. Ultimately will enhance patient care, and relief stress on GPs. Ensuring the sustainability of the NHS workforce.

In Appendix 6, 'The Number of GPs Trend,' provides a complex picture of recent data for the general practitioner (GP) workforce in Scotland. Since 2017, there were another 271 GPs joining the workforce in Scotland, leading to the overall headcount of individual GPs rising and to the majority of practices in Scotland having an average of six GPs per practice. Despite this, the Scottish Government is likely to fall short of reaching its target of increasing the total GP headcount by 800 by 2027 (BBC News, 2017).

According to the FTE figures the fall in full-time equivalent (FTE) fully qualified GPs was revealed. It's the number of hours worked by a GP as if it were a full-time (FT) job. Headcount is rising, but overall, the number of FTE GPs has fallen by 197 compared to 2013, or almost 5%. The mismatch between the number of actual GPs working full-time, versus the number required to fill full-time, stable positions, suggests that many GPs may be working fewer hours, working part-time, or are being assigned locum roles to fill rather than stable full-time positions.

Furthermore, the number of performing GP partners has decreased particularly sharply. GP partners are senior GPs who own and are responsible for management and clinical work in practices. But, since 2010, the number of GP partners has plummeted by 628 to 3,151 in 2023. This is a trend that places practice management in the hands of long-term experienced GPs to meet the demands of the practice but carries the challenge of not always being able to retain them.

They are also playing catch up with GP practices in Scotland that have high vacancy rates. In 2022/23 the 42.3% of GP practices who reported at least one unfilled GP position represent 1.3 million patients. The high vacancy rates place extra workloads on the existing GPs and increase waiting times for patients, and there is a critical need for strategies to attract and retain GPs throughout Scotland.

The UK's General Medical Council (GMC) has 390,049 doctors on its register, but just 21,255 new doctors were granted registration for the first time last year, a sign of lagging numbers of professionals entering the workforce. Obviously rigorous training and quality standards are very important to the delivery of competent healthcare, but I think an excessive length of these qualifications limits the number of fresh healthcare doctors available to meet increasing needs and invariably, these are often in areas such as emergency medicine. Scotland needs more doctors. This is a problem worth solving quickly in Scotland, so it can attract doctors from abroad. That should help fill the shortage without harming quality of care or damaging the strong Scottish healthcare system, said Scott Tomblin, a professor of physiotherapy at Abertay University, which includes Dundee Medical School.

2.1.1 Recommendation

Opening Opportunities Internationally

To mitigate the issue of a shortage of doctors within the country and emergency medicine in particular in Scotland several strategies involve the acquisition of doctors from other countries. Since CHBs cooperate with other global health boards, the Scottish healthcare system plans to enhance the allowance of practicing licenses thus minimizing bureaucratic procedures for Jinping-qualified doctors. Furthermore, recruitment drives will be employed to spotlight the available options for doctors from other countries most especially for those willing to come and work in the UK in specialized departments such as the A & E department. Scots and international students studying medicine, with the possibility to focus on emergency care, are offered scholarships, which will also contribute to the development of the personnel base. Approving visas and providing doctors with clearly defined visa upgrades from less developed countries prolongs the time doctors will spend in Scotland.

Financial Incentives for Junior Doctors and Interns

From the financial management perspective, competition remains a crucial determinant of the financial remunerations necessary to lure and maintain such human capital. Junior doctors are paid an annual wage that lies in between £29, 000 and £40, 000, which may be considered as a good starting pack for young career chasers. The first scheme is creating internship, where 50 percent of the junior doctor salary is to be paid to the intern and an additional completion bonus, helping new entrants to integrate the medical field. Moreover, loan forgiveness program is aimed to help the international students and junior doctors to avoid the debts but on the condition that the individuals will have to serve for three to five years at least. The scheme links subsidized training to workforce sustainability so young doctors will put their early working years into The Scottish health services.

2.1.2 Outcome

All these changes are expected to bring about improvements in the Scottish health care systems. That is, positive relations between the shortage of skilled human resources and the quality of healthcare services will be established; negative relations between the shortage of skilled human resources and patient waiting time will be ascertained; and correlations between the shortage of skilled human resources and junior doctors support, and overall patient satisfaction will be affirmed. Having more people employed will also work to reduce the current high turnover that is being experienced among health care providers. The effectiveness of such measures in the long term is projected to cause a lessening of the burden on public health service and further decline on overall costs and increase in people's health in Scotland.

2.2 Quality of care

The quality of care in Accident and Emergency departments is a vital issue in the delivery of health care because it impacts outcome measures and the perception of a health care system. It is so because we have highlighted the quality because of its impact on the patients and the healthcare providers; both are affected by the

impressions and belief in the system in healthcare or by the perished because of the tiredness and burnout of the growing number of patients. Between 7 September and 30 October 2022, a total of 3,362 participants were asked how satisfied they were overall satisfaction with the National Health Service (NHS) and the results came out to be a huge drop in the satisfaction rate of the National Health Service (NHS) published during 2022. Overall satisfaction with the National Health Service (NHS) is down to 29 percent, 7 percent points in 2021, the survey shows. The survey found satisfaction at this level is the lowest since the survey began in 1983. Also found to be more satisfied with the NHS than the country overall has been more than half (51 per cent), the highest figure in the survey's history. The People surveyed had three reasons from which they got to select the reasons to be happy and the reasons to be unhappy with the National Health Service (NHS). The top three reasons people are not satisfied are not being able to see their general practitioner and hospital appointments waiting times and a perception that the government doesn't spend enough on the National Health Service (NHS). The highest reasons for satisfaction with care reasons were free at point of use (74 per cent), quality of care (55 per cent), and a good range of services and treatment available (49 per cent).

2.2.1 Recommendation

Proposed System to Capture Real-Time Patients' Experience

We propose a system to capture the journey of patients in the National Health Service (NHS) whether the patient have requested an ambulance or went by themselves to the hospital, we want them to be able to express themselves and make them feel heard, supporting the values of the NHS of *inclusivity* and *responsiveness* of patient needs.

2.2.2 Outcome

Our vision is to place the application in Accident and Emergency (A&E) departments, allowing it to capture patients' arrival and wait times, record ambulance wait times, and gather feedback on ambulance to ward or room. Patients would also be able to provide feedback throughout their stay, including ambulance experiences, interactions with Accident and Emergency (A&E) staff, and during admission and discharge. We would also ease the process with our mobile interface. The application would also display staff names and faces for easy identification and allow open-ended suggestions for improvement, in line with the National Health Service (NHS) goals for patient-centered care and continuous improvement.

3.0 Limitations

3.1 Insufficient Staffing Level

However, these strategies are surrounded by the following difficulties. The players that need to be licensed are often encased in complex regulations to reduce the rate at which foreign doctors are recruited hence, members. Also, they include scholarship programs, visa sponsorships, and loan forgiveness, which demographically and

geographically require huge funding hence likely to encounter financial hitches. Retention is another concern still: some international doctors may be viewed as having adaptation concerns that cord on their ability to remain long term. To overcome these limitations, appropriate solutions during onboarding processes and regarding personal finances would be required to use these recruitment and retention strategies most effectively.

3.2 Quality of care

However, using a real-time patient feedback app in healthcare is a challenge especially in busy places such as the A&E ward. That means not everyone will want to use the app and some people just don't like giving feedback in that way. Staff may not buy in if the app doesn't accurately reflect what goes down during patient care, after all, and they may also not be fully on board if they feel such a high level of assurance isn't possible.

It needs to be trained for staff and patients to get the app to work well. It is also very important to follow the rules and protect patient privacy building trust. People have to support the app and analyse the data as well as follow up on the feedback which costs money. These could cause the app to not perform well, or even worse to not work at all for long periods.

4.0 Expected Benefits in Practice

We believe by implementing the system, the National Health Service (NHS) could have many positive outcomes from it, we list the benefits in table in Appendix 4 with benefits, expected impacts, real-world examples, and specific areas from each of the United Kingdom (UK) countries (Scotland, England, Wales, and Northern Ireland). These examples illustrate how National Health Service (NHS) trusts across the United Kingdom (UK) are actively implementing programs and moving towards more digitalisation in the health sector.

5.0 Conclusion

Despite these challenges, prioritizing the interests of patients and fostering a positive relationship between patients and NHS staff can lead to meaningful improvements in care quality. By integrating such a feedback application into A&E, we aim to enhance both patient experience and trust in healthcare services, ultimately supporting NHS goals in delivering compassionate, responsive care.

Future analyses for the proposed patient feedback application in A&E departments will focus on several key areas. First, a longitudinal analysis will track patient feedback over time to identify trends and seasonal variations in experiences, while comparative analyses will evaluate feedback across different departments to highlight best practices and areas needing improvement (Topol, 2019). Additionally,

researchers will investigate the correlation between patient feedback scores and clinical outcomes, such as readmission rates, as well as how patient satisfaction relates to clinical performance metrics (Darzi & Wachter, 2020). Demographic analyses will examine feedback based on patient characteristics to identify disparities and ensure equitable care (Healthwatch, 2018), alongside an assessment of technology access and usage to identify barriers to engagement. Furthermore, the application of individual staff evaluations using patient feedback will facilitate recognition and training opportunities, while exploring team dynamics will help understand their influence on patient feedback scores (National Institute for Health Research, 2018). Follow-up analyses will assess the impact of feedback on subsequent patient experiences, as well as measure the effectiveness of actions taken in response to the feedback (Tebra, 2023). The adoption rates of the feedback application will be tracked to identify engagement barriers, and the effectiveness of the feedback loop will be evaluated by examining staff responsiveness. Data privacy assessments will focus on compliance with regulations to protect patient information (General Data Protection Regulation [GDPR], 2016), alongside surveys to gauge patient trust in the feedback system. Finally, a cost-benefit analysis will evaluate the financial implications of the feedback application against the benefits of improved patient satisfaction and resource allocation efficiency (Darzi & Wachter, 2020). Collectively, these analyses will provide valuable insights into the effectiveness of the patient feedback application, supporting continuous improvement in patient experiences and care quality while enabling the NHS to make data-driven decisions.

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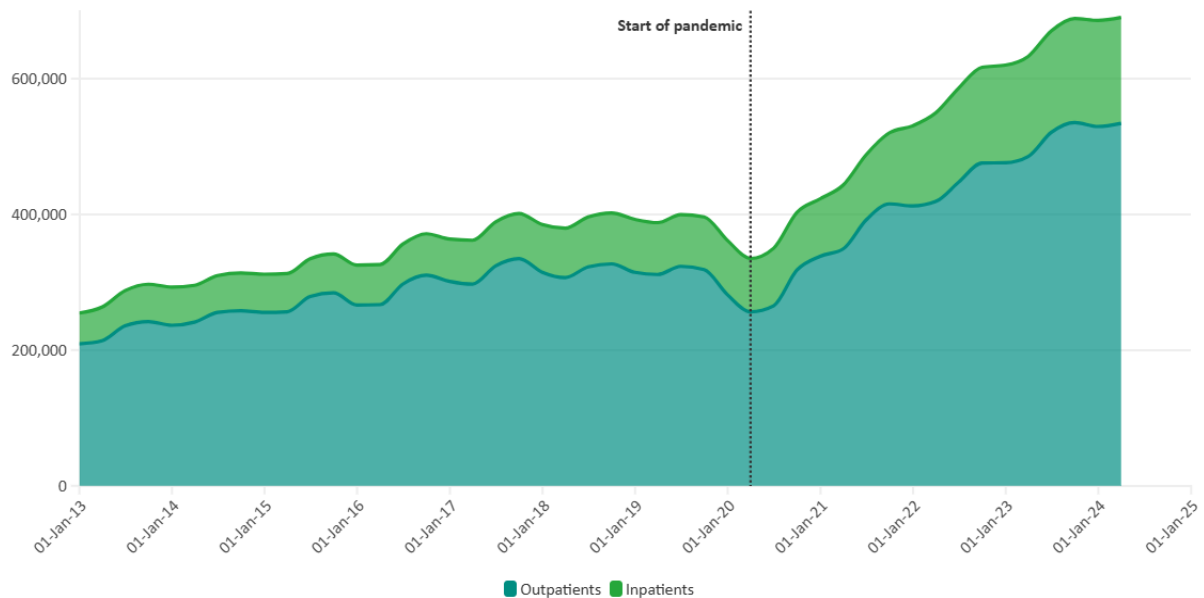
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Appendices

Appendix 1: Number of Patients waiting for Treatment

January 2013 - March 2024

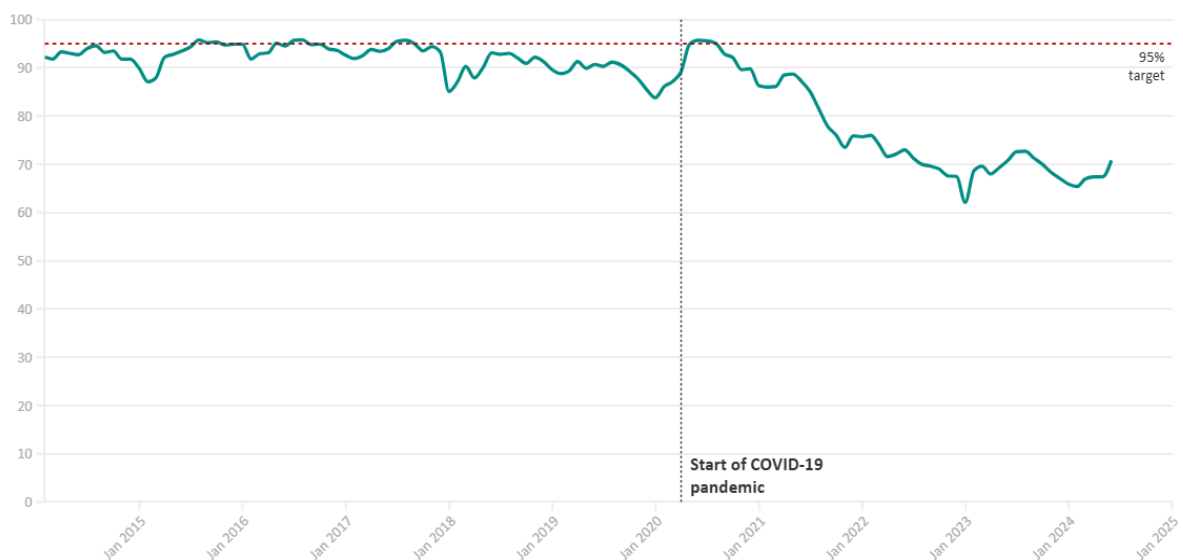


Source: BMA analysis of [Public Health Scotland NHS Waiting Times data](#)



Appendix 2: Waiting times in A&E

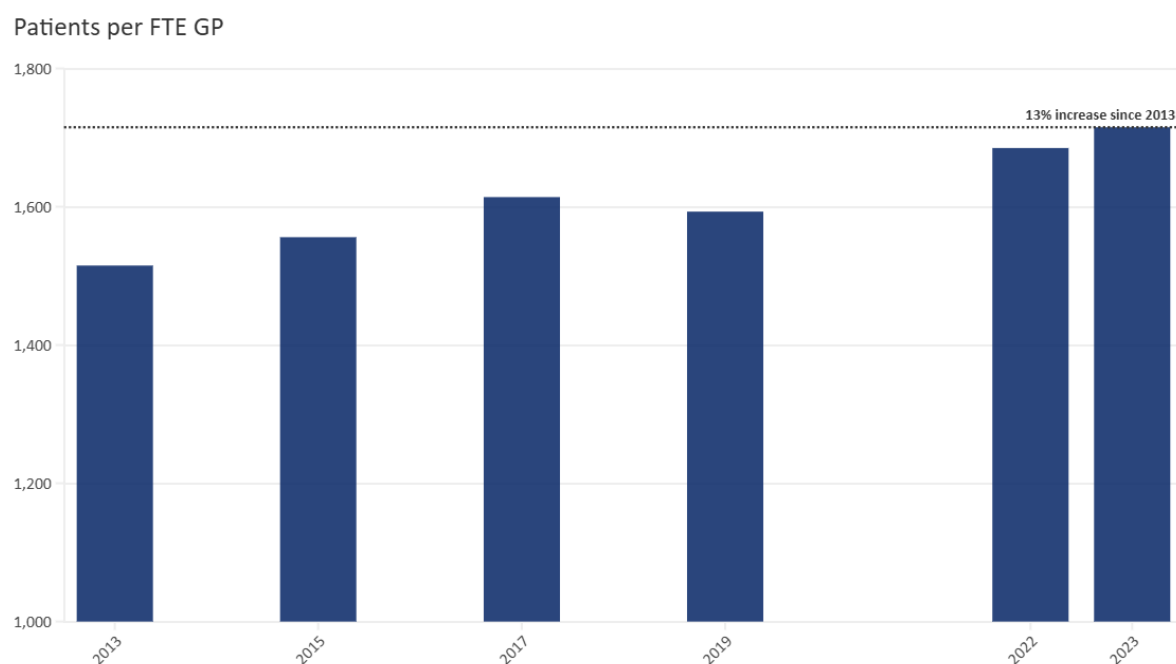
Patients seen within 4 hours (%)



Source: BMA analysis of [Public Health Scotland data](#)



Appendix 3: Growing Patient Load per GP



Appendix 4: Incorporating Patient Experience Feedback.

Performance monitoring and assurance	Shared understanding and information	Improvement
Comparison with other healthcare providers	Helping people to make choices about services	Improvement and redesign of services
Monitoring impact of service changes	Understanding problems in services	Reflection on healthcare professionals' behaviours
Informing commissioning decisions	Public accountability	Frame care as person-centred rather than task or outcomes based
Compliance with standards	Increasing healthcare professionals understanding of the patients' real life experience	Co-designing services with staff and patients

(Source: National Institute for Health Research, 2021)

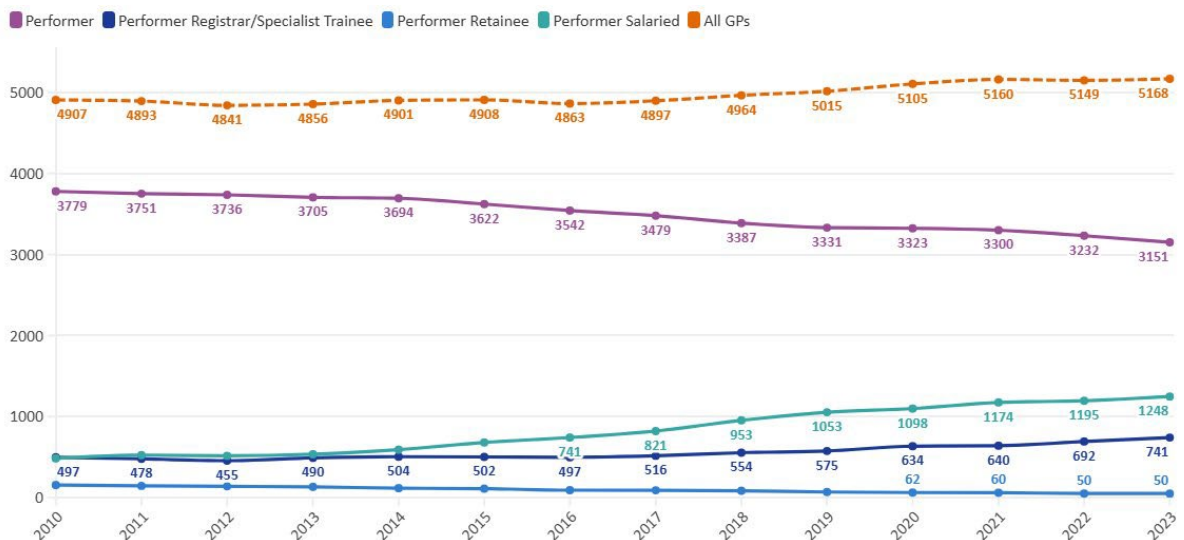
Appendix 5: Benefits and Regional Examples of NHS Improvements

Benefit	Expected Impact	NHS Real-World Example	Country and Area
	Faster recovery with fewer complications	The <i>Enhanced Recovery after Surgery</i>	

Improved Patient Outcomes		(ERAS) program at NHS Lothian focuses on faster access to diagnostics and rehabilitation, which has improved recovery times for surgical patients.	Scotland, NHS Lothian
		The <i>National Enhanced Recovery After Surgery (ERAS) Programme</i> in Wales optimizes perioperative care, leading to quicker recoveries and fewer complications.	Wales, National Programme
Reduced Waiting Times	Efficient processes for timely treatment	The NHS Improvement initiative introduced the <i>Same Day Emergency Care</i> (SDEC) model to reduce wait times by offering assessments and treatments on the same day, whenever possible.	England, NHS Improvement
		The <i>1000 Lives Improvement</i> initiative in Wales reduces waiting times by streamlining patient flow.	Wales, National Initiative
Improved Patient Satisfaction	Enhanced feelings of being valued	<i>Patient and Client Experience Standards</i> in Northern Ireland ensure patients feel respected and valued.	Northern Ireland, Regional Implementation
		The <i>Patient First</i> program at Brighton and Sussex University Hospitals enhances patient satisfaction through feedback-driven improvements.	England, Brighton and Sussex
Enhanced Staff Well-being & Retention	Boost in morale and workforce stability	<i>Health and Well-being Support Framework</i> in NHS Wales offers mental health resources to reduce burnout and support retention.	Wales, National Framework
		<i>Staff Well-being Hub</i> at East Lancashire Hospitals NHS Trust provides counseling and wellness programs, improving retention.	England, East Lancashire

Increased Efficiency & Cost Savings	Reduced costs and minimized delays	The <i>Value-Based Healthcare Programme</i> in NHS Wales prioritizes outcomes that matter to patients while using resources efficiently.	Wales, National Programme
		<i>Quality 2020</i> in Northern Ireland improves efficiency by managing resources effectively and delivering high-quality care.	Northern Ireland, Regional Strategy
Prevent Loss of Important Patient Documents and Files	Reliable access to patient records ensures crucial information is retained and accessible	The <i>Welsh Clinical Portal</i> offers a unified digital platform for storing patient records, reducing the risk of lost documents.	Wales, National Implementation
		<i>EPIC electronic health record system</i> at Guy's and St Thomas' NHS Foundation Trust in London secures and organizes patient files.	England, London

Appendix 6: The Number of GPs trend



Source: NHS Education for Scotland • Performer: a GP who has entered into a contract to provide services to patients and is effectively self-employed; usually a practice partner. Performer salaried: A GP who is employed by the practice or NHS Board on a salaried basis. Performer retainee: A GP, typically part-time, who can be utilised by a practice as required. Performer Registrar/Specialist Trainee: A medical practitioner in a GP training program. As some GPs may hold multiple posts simultaneously, the total number of GPs ('All GPs') may not equal the sum of GPs across each designation.